Networking.

A very important component of any lab environment is understanding how networking works. This is important so that we understand how the devices communicate with each other. I will use a series of analogies to make this concept easier to understand; keep in mind that this will cover most of the topics we discuss in the class, but does not cover all of the networking nuances.

The components we will be talking about are

- IP address
- Port
- NAT
- Subnet Mask
- Network address
- Host address
- Broadcast address
- Router
- DHCP and Static IP
- DNS
- Switch
- Route
- Public and Private IP addresses
- Default Gateway
- Bits
- Network interface (NIC)

IP address – Just as a house address, this designates the exact location where packets are going.

As you have probably seen and IP address is composed of four octets, each separated by a period. Ex.

- 192.168.1.10
- 10.10.10.10
- 15.255.30.10

Now one thing you might be thinking "I thought that computers operate on '0's and '1's (bits), how does this fit in. Good question. The representations above are actually created to make the IP addresses easier for humans, computers translate those into a string of bits. An IP address is made of 32 bits. So, in case you were wondering (192.168.1.10 = 11000000101010000000000100001010). If you are curious, you can easily find articles on how to convert decimal to binary (bits) and it's not needed for our use cases, so I won't go over it in more detail here.

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Remember, bit is a 0 or a 1 used by computers. Binary means we are representing numbers in a sequence of 0s and 1s.

Network address – this is the address of the network a device is on, think of it as a neighborhood.

Host address – this is the portion of the IP address that represents the host; think of this as the house number in the neighborhood.

Great. But how in the world does the computer know which portion is the network address and which part is the host portion. This is where the Subnet Mask comes in.

Subnet Mask – this is the missing piece that tells the computer which part is which. There are usually two types of representations that you'll see

- 255.255.255.0 – in Decimal format
- /24 – slash notation.

How does IP address and Subnet Mask work together you ask? It works based on the Boolean Logic "AND". Boolean Logic is done on binary data; yup the 1s and 0s I mentioned you won't have to worry about converting. The only thing you have to know is that in a subnet mask, 255 = 11111111 (eight ones in binary) and the 0 = 00000000 (eight zeros). Remember an IP address is 32 bits long, that's why the 0 must be converted to the longer notation. So, when you look at 255.255.255.0 you have 24 ones followed by 8 zeroes. Does 24 sound familiar? I certainly hope so, see the two examples above (255.255.255.0 = /24).

Since we are only dealing with octets where we have either a 255 or 0, our job of separating the network address from the host address is pretty simple. Align the IP and subnet on top of each other; a 255 means that piece belongs to the network address and a 0 means it belongs to the host portion.

So, in our example IP of 192.168.1.10 with a mask of /24, we would do the following

192 168 1 10
255 255 255 0

Network portion: 192.168.1
Host address: 10
(Network address is 192.168.1.0, we add the 0 to make it 32 bits in length).

Why do we need to know this? Well, usually devices that are on the same network/neighborhood can easily communicate with each other. Going to other networks usually requires some sort of intermediary device such as a router. Keep this in mind when you are looking at your diagram and configuring your networks.

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Here are a few for you to try (solutions at the end of document)
10.23.13.5/16
80.15.45.4/8
55.41.31.77/24

Let's quickly touch on a router, switch, default gateway, route, NIC, broadcast address.

Router is a device that moves packets from one network to another and usually has information about many other networks that are not necessarily directly connected to each other. Think of going to any website. You have to go through multiple routers to get to the address of the website.

Switch is a device that historically allowed multiple devices to connect to the same network.

Default gateway (DG) tells a device where to send packets if the device doesn't know where they should go. Usually, the DG has the IP that is one above network address ex. 192.168.1.1. To communicate, we usually need to send packets to a different network and that's why most time a device such as a router, which has knowledge of different networks is specified as the default gateway.

A route is the road that packets will travel to get to destinations. There are dynamic and static routes. Dynamic routes are usually self-learned by the device and are other devices in the same network. Static routes on the other hand are routes that we as administrators manually enter into devices to tell the packets exactly where they should go and out of which NIC.

NIC is the network adapter used to send and receive information from the network. In many of our machines, we will attach multiple NICs on different networks.

Broadcast address is a special type IP address which is used by a device when it doesn't know routes and IPs of other devices on the network. Usually, the IP address ends in .255.

The final piece we should discuss is ports and NAT.

Imagine you have multiple software applications running on a server; how would the server know which application you are trying to interact with? You could technically have an IP for each service, but you would quickly run out of IPv4 addresses this way. This is where ports come in. Ports make it easier for the server to differentiate where the traffic is supposed to be going. You have probably heard about the below port/service combinations

- http – 80
- https – 443
- SSH – 22
- SMTP 25

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As you may have heard, there is a limited supply of IPv4 addresses. In the world, there is a total 4,294,967,296 addresses. When the designers were thinking about this number in the past, they never imaged that such a huge number would ever be used. However, as the internet was developing, it was pretty obvious that this number would not be enough. Hence, they created NAT – Network Address Translation. NAT allows for the use of 1 IP address to function as the address for thousands of devices. When a device behind NAT wants to communicate with the outside world, it reaches out to the NAT device (usually a router). The router then attaches a port number to that device and sends it along. When traffic targeted at the devices comes back, the router sees at which port the information is destined to and based on this sends it off to the correct device. In our example, imagine an apartment building having one address. Each apartment would have the same building address, but its own apartment number (port).

Congratulations, you have just learned the basics of networking that will be used throughout. Make sure you have this document handy as you are configuring your devices. Let me know if you have any questions.

Solutions

10.23.13.5/16

- Network portion 10.23
- Host portion 13.5

80.15.45.4/8

- Network portion 80
- Host portion 15.45.4

55.41.31.77/24

- Network portion 55.41.31
- Host portion 77

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